BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Paper Physics - 11th — Annual Exam – 2021 — SECTION- A

Time Allowed: 20 mints | **Maximum Marks:** 18

Note: Q.1: Fill in the blanks with sutable option.

Q1. The longer the pendulum, greater will be it's.....

(A) Frequency
(B) Time period
(C) mass
(D) None

Q2. The time period of a simple pendulum is 2 sec. It its length is increased by 4 times then its T.....

(A) 8sec
(B) 12sec
(C) 4sec
(D) 16sec

Q3. The portion of a wave below the mean position is called...

(A) Crest
(B) trough
(C) wave length
(D) amplitude.

Q4. The principle of diffraction grating is based on the .....and diffraction of light waves.

(A) Reflection
(B) refraction
(C) interference
(D) none

Q5. Maximum work can be obtained in .....process.

(A) Adiabatic
(B) Isochoric
(C) cyclic
(D) Isothermal

Q6. 2π radians = ...

(A) 360^0
(B) 1 revolution
(C) 180^0
(D) Both A and B

Q7. Work can be expressed in terms of base units are.....

(A) Kg ms^{-2}
(B) kg $m^2.s^{-2}$
(C) kg $m^2.s^{-3}$
(D) kg $m^{-1}.s^{-2}$

Q8. The end point is usually called .....of the vector.

(A) Tail
(B) origin
(C) mid
(D) head

Q9. When moment arm "r"and force are parallel then Torque is.....

(A) Negative
(B) maximum
(C) zero
(D) minimum

Q10. The vertical component of velocity changes uniformly and is.....at height point in projectile motion.

(A) Maximum
(B) minimum
(C) constant
(D) zero

Q11. If $\theta = 1800$ then work done is.....

(A) Positive
(B) zero
(C) negative
(D) maximum

Q12. In British Engineering system the unit of power is...

(A) Pascal
(B) Joule
(C) Watt
(D) horse power

Q13. If momentum is increased by 20% then kinetic energy increases by.....

(A) 55%
(B) 44%
(C) 77%
(D) 66%

Q14. The minimum required velocity to put a satallte inot the orbit is called.....velocity.

(A) Cirtical
(B) terminal
(C) angular
(D) escape

Q15. Moment of Inertia of a Ring or Thin walled cylinder is I ...

(A) $1/2MR^2$
(B) MR^2
(C) $2/5MR^2$
(D) $1/12MR^2$

Q16. The unit for a solid angle in three deimensional space is...

(A) Radian
(B) steradian
(C) Candela
(D) Revolution

Q17. Bernoulli's equation is based upon law of conservation of..

(A) Momentum
(B) charge
(C) Mass
(D) energy

Q18. Density of air is.....

(A) 1.2kg. m^{-3}
(B) 1.2g. cm^{-3}
(C) 11.2 $kg.m^{-3}$
(D) 11.2 g. cm^{-3}

SECTION-B

Time Allowed: 2.40 Hrs. | **Maximum Marks:** 40

Q2. (Marks 40) Q.2: Attempt any ten (10) parts. All parts carry equal marks.

i. Show that rate of change of angular momentum is equal to torque.

ii. Define terminal velocity and prove that $V_t \propto r^2$.

iii. What will be the frequency of a simple pendulum if its length is 1m?

iv. Define and briefly explain interference of sound waves.

v. State and explain Breweste's law of polarization.

vi. The pressure in a gas containing hydrogen will leak more quickly than if it is containing oxygen. Why?

vii. Define error and its types.

viii. What is the minimum number of unequal vectors to result into a null vector? Explain with the help of diagram.

ix. State and explain law of conservation of momentum.

x. Define projectile motion. Derive mathematical form for maximum height attained by projectile.

xi. Show that work done along a closed path is zero.

xii. Define and briefly explain vector product of two vectors.

xiii. Show that $E = mc^2$ is dimensionally correct.

SECTION-C

Maximum Marks: 27

Note: Note: Attempt any Three of the following questions. All questions carry equal marks.

Q3.

a) Explain Laplace's correction in theoretical and experimental values of sound in gas.

b) Light is incident normally on a grating which has 350 lines/mm. Find the wave length of spectral line for which the deviation in second ordre is 25°.

Q4.

a) Define the molar heat capacities C_p and C_v for a gas. Show that for a mole of an ideal Gas C_p-C_v = R.

b) A 5gm mass at the end of a spring describes S.H.M with a period of 0.08 sec. Find the acceleration when displacement is 6 cm.

Q5.

a) State he equation of continuity. Show that $V \propto 1/A$ where 'V'is the speed of fluid and A is cross sectional area of the pipe.

b) An 80 kg Athlete runs upstairs in 6 sec. the vertical height of the stairs is 5480 cm. Calculate his power.

Q6.

a) Define centripetal acceleration. Show that $ac = -\omega^2 r$

b) A person on the sea shore observes that 68 waves reach the shore in 1 min. If the wave length is 12, find velocity of waves.